

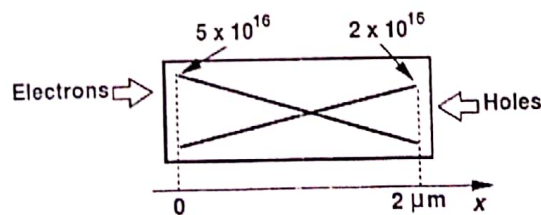
DEPARTMENT OF ELECTRONICS ENGINEERING
Indian Institute of Technology (Indian School of Mines), Dhanbad
Dhanbad, Jharkhand- 826004, India

SUBJECT: ELECTRONICS ENGINEERING (ECC 12101)

CLASS: B. TECH 2ND SEMESTER, SEC.-A,B,C,D

ASSIGNMENT I

1. Determine the probability that an energy level $3kT$ above the Fermi energy is occupied by an electron. Let $T = 300$ K. 0.0474
2. A Si sample is doped with 10^{17} As atoms/cm³. What is the equilibrium hole concentration p_0 at 300 K? Where is E_F relative to E_i ? ($n_i = 9.65 \times 10^9$) 931.225 atoms/cm³
3. An n-type silicon bar 0.1 m long and $100 \mu\text{m}^2$ in cross-sectional area has a majority carrier concentration of $5 \times 10^{20} / \text{m}^3$ and the carrier mobility is $0.13 \text{ m}^2 / \text{V-s}$ at 300K. If the charge of an electron is 1.6×10^{-19} coulomb, then calculate the resistance of the bar. 96.1538 Ω
4. Determine the density of electrons in silicon at $T = 300$ K (room temperature) and $T = 600$ K. [$E_g = 1.12$ eV, $B = (5.2 \times 10^{15})^2$] 1.526 $\times 10^{15}$
5. A uniform piece of n-type of silicon that is $1 \mu\text{m}$ long senses a voltage of 1 V. Determine the velocity of the electrons. ~~1120 A/cm²~~ 13×10^8
6. The electron concentration in a sample of uniformly doped n-type silicon at 300K varies linearly from $10^{17} / \text{cm}^3$ at $x=0$ to $6 \times 10^{16} / \text{cm}^3$ at $x = 2 \mu\text{m}$. Assume a situation that electrons are supplied to keep this concentration gradient constant with time. If electronic charge is 1.6×10^{-19} coulomb and the diffusion constant $D_n = 35 \text{ cm}^2 / \text{s}$, calculate the current density in the silicon, if no electric field is present. 1120 A/cm²
7. The resistivity of a uniformly doped n-type silicon sampled is $0.5 \Omega\text{-cm}$. If the electron mobility (μ_n) is $1250 \text{ cm}^2 / \text{V-sec}$ and the charge of an electron is 1.6×10^{-19} coulomb, calculate the donor impurity concentration (N_D) in the sample. 10^{22} atoms/cm³
8. The mobilities of electron and hole in silicon are respectively $1350 \frac{\text{cm}^2}{\text{V-sec}}$ and $480 \frac{\text{cm}^2}{\text{V-sec}}$ and intrinsic carrier concentration of silicon is $1.5 \times 10^{10} / \text{cm}^3$. Then find the majority and minority concentrations of a p-type semiconductor which is having resistivity $10 \Omega\text{-cm}$. 1.7578×10^5
9. The resistance of number 18 copper wire (diameter 1.03 mm) is $0.0214 \Omega / \text{m}$. The concentration of free electron is 8.4×10^{28} electron/cm³. If it is carrying a current of 2A. Find drift velocity and mobility of electron. 4.17×10^{-9}
10. Figure shows a p-type bar of Silicon that is subjected to electron injection from the left and hole injection from the right. Determine the total current flowing through the device if the cross-section area is $1 \mu\text{m} \times 1 \mu\text{m}$. 0.1144 μA



11. A n-type piece of Silicon with a length of $0.1 \mu\text{m}$ and a cross section area of $0.05 \mu\text{m} \times 0.05 \mu\text{m}$ sustains a voltage difference of 1 V. If the doping level is 10^{17} cm^{-3} , calculate the total current flowing through the device at $T = 300$ K. 0.52 μA

Assignment I

given:- $E - E_F = 3kT$

$T = 300\text{ K}$

$$F(E) = \frac{1}{1 + e^{\frac{(E - E_F)}{kT}}} = \frac{1}{1 + e^{\frac{3kT}{kT}}} = \frac{1}{1 + e^3}$$

$$F(E) = 0.0474$$

2) given:- $n = 10^{17} \text{ atoms/cm}^3$

$T = 300\text{ K}$

$n_i = 9.65 \times 10^9$

(i) $p_o = ?$

$$n \times p = n_i^2$$

$$p = \frac{n_i^2}{n} = \frac{(9.65 \times 10^9)^2}{10^{17}} = 931.225 \text{ atoms/cm}^3$$

(ii) E_F is relatively above E_i due to heavily doped n type material.

$$\text{Shift} = kT \log_e \left(\frac{N_D}{n_i} \right) \text{ eV}$$

$$= 1.38 \times 10^{-23} \times 300 \times \log_e \left(\frac{10^{17}}{9.65 \times 10^9} \right)$$

$$\text{Shift} = 6.6876 \times 10^{-20} \text{ eV}$$

3) given:-

$$l = 0.1 \text{ m}$$

$$A = 100 \mu\text{m}^2$$

$$N_D = 1.6 \times 10^{19} \text{ C}$$

$$N_D = 5 \times 10^{20} \text{ /m}^3$$

$$\mu_n = 0.13 \text{ m}^2/\text{V-s}$$

$$T = 300 \text{ K}$$

$$\Rightarrow R = \frac{\rho l}{A} \quad , \quad \rho = \frac{1}{\sigma} \quad , \quad \sigma = q n \mu$$

$$\sigma = 1.6 \times 10^{-19} \times 5 \times 10^{20} \times 0.13$$

$$\sigma = \underline{10.4 \text{ V/m}} \quad \text{or} \quad 10.4 \text{ m}^2/\text{V-sec}$$

$$\rho = \frac{1}{\sigma} = 0.09615$$

$$R = \frac{0.09615 \times 0.1}{100 \times 10^{-6}} = \underline{96.1538 \Omega}$$

4) given :- $E_g = 1.12 \text{ eV}$

$$B = (5.2 \times 10^{15})^2$$

(i) $T = 300 \text{ K}$

$$n_i = B_0 \cdot T^{3/2} \cdot e^{-\frac{E_g}{2kT}}$$

$$n_i = (5.2 \times 10^{15}) \times (300)^{3/2} \times e^{-\frac{1.12}{2 \times 8.6 \times 10^{-5} \times 300}}$$

$$\boxed{n_i = 1.077 \times 10^{10}}$$

4) $T = 600K$

$$n_i = B^{1/2} T^{3/2} e^{\frac{-E_g}{2kT}}$$

$$= 5.2 \times 10^{15} \times (600)^{3/2} \times e^{\frac{-1.12}{2 \times 1.38 \times 10^{-23} \times 600}}$$

$$= 5.2 \times 10^{15} \times (600)^{3/2} \times e^{\frac{-1.12}{1.6 \times 10^{-19}}}$$

$$n_i = 1.526 \times 10^{15}$$

5) given:- $l = 1 \times 10^{-6} m$

$$V = 1V$$

$$\mu_{n-si} = 1300$$

$$V_d = \mu E = \mu \times \frac{V}{l}$$

$$= 1300 \times \frac{1}{1 \times 10^{-6}}$$

$$V_d = 13 \times 10^8$$

6) $J = q D_n \cdot \frac{dn}{dx}$

$$= 1.6 \times 10^{-19} \times 35 \times \left[\frac{N_D|_{x=0} - N_D|_{x=2}}{0 - 2 \times 10^{-4}} \right]$$

$$\Rightarrow 1.6 \times 10^{-19} \times 35 \left[\frac{10^{17} - 6 \times 10^{16}}{2 \times 10^{-4}} \right]$$

$$\Rightarrow -1120 A/cm^2$$

$$J = -1120 A/cm^2$$

$$\textcircled{7} \quad \beta = \frac{1}{\sigma} = 0.5 \Omega \text{ cm}$$

$$\text{given } \mu n = 1250 \text{ cm}^2/\text{V} \cdot \text{sec}$$

$$\text{charge } = e = 1.6 \times 10^{-19} \text{ C}$$

To find $\rightarrow$ $N_D = ?$. Donor impurity concentration = ?

$$\text{Sol} \rightarrow \beta = \frac{1}{\sigma} \quad \sigma = q \mu n$$

$$\beta = \frac{1}{q \mu n}$$

$$0.5 \times 10^{-2} = \frac{1}{1.6 \times 10^{-19} \times 1250 \times 10^4 \times n}$$

$$n = \frac{1}{1.6 \times 10^{-19} \times 1250 \times 10^4 \times 0.5 \times 10^{-2}}$$

$$= \frac{10^{25}}{1.6 \times 1250 \times 0.5}$$

$$= \frac{10000000 \times 10^{19}}{1.6 \times 1250 \times 0.5} = 10^{22} \frac{\text{atom}}{\text{cm}^3}$$

atom/m³

given

$$\textcircled{8} \quad \mu_n = 1350 \frac{\text{cm}^2}{\text{V-sec}} \quad \mu_p = 480 \frac{\text{cm}^2}{\text{V-sec}}$$

$$n_i = 1.5 \times 10^{10} / \text{cm}^3 \quad \beta = 10 \Omega\text{-cm}$$

to find : minority concentration ?
majority conⁿ ?

$$\rightarrow \beta = \frac{1}{\sigma}$$

$$\sigma = q \left(\mu_n \frac{n_i^2}{P} + P \mu_p \right)$$

$P \Rightarrow$ majority conⁿ $P_n = \frac{n_i^2}{P}$
 $\frac{n_i^2}{P}$ minority conⁿ

$$\frac{1}{10} = 1.6 \times 10^{-19} \left(1350 \times \frac{2.25 \times 10^{20}}{P} + P \times 480 \right)$$

$$n_i^2 = (1.5 \times 10^{10})^2 = 2.25 \times 10^{20}$$

$$\frac{P}{10 \times 1.6 \times 10^{-19}} = 1350 \times 2.25 \times 10^{20} + P^2 \times 480$$

$$\frac{P \times 10^{19}}{16} = 1350 \times 2.25 \times 10^{20} + P^2 \times 480$$

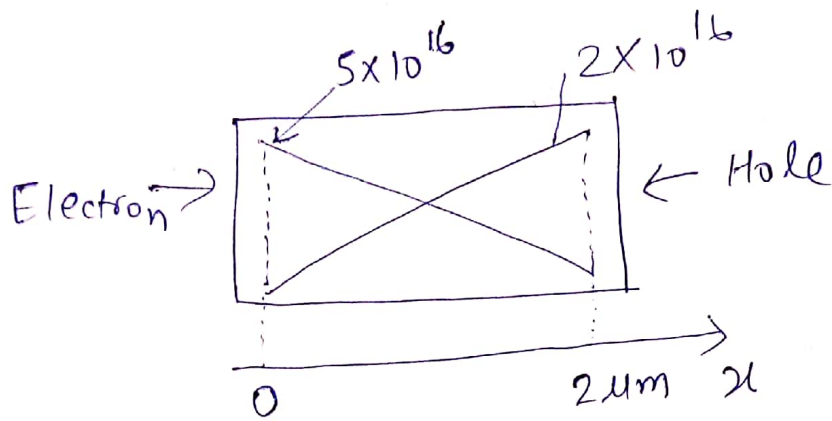
$$P \times 10^{19} = 1350 \times 16 \times 2.25 \times 10^{20} + P^2 \times 16 \times 480$$

$$P^2 \times 16 \times 480 - P \times 10^{19} - 1350 \times 16 \times 2.25 \times 10^{20} = 0$$

$$P = 10^{15} \times 1.28 \text{ w/cm}^3$$

$$P_n = \frac{n_i^2}{P} = 1.7578 \times 10^5 \text{ cm}^{-3}$$

Area = 10^{-12} m^2



To find total current

$$I_n = (qnq \frac{dn}{dx}) A$$

$$I_n = nqA \frac{5 \times 10^{16}}{2 \times 10^{-6}}$$

$$I_p = -p q A \frac{dp}{dx}$$

$$I_p = -p q A \times \frac{2 \times 10^{16}}{2 \times 10^{-6}}$$

$$I_T = I_n + I_p$$

$$\frac{pn}{\mu n} = V_T$$

$$\frac{pn}{1300} = 26 \times 10^{-3}$$

$$pn = 13 \times 26 = 33.8$$

$$\frac{pP}{\mu p} = V_T$$

$$\frac{pP}{500} = 26 \times 10^{-3}$$

$$pP = 26 \times 500 \times 10^{-3} = 26 \times 0.5 = 13$$

$$I_n = 33.8 \times 1.6 \times 10^{-19} \times 10^{-12} \times \frac{5 \times 10^{16}}{2 \times 10^{-6}} = \frac{33.8 \times 1.6 \times 5}{2} \times 10^{-9} = 1352 \times 10^{-9} \text{ A}$$

$$I_p = -26 \times 1.6 \times 10^{-19} \times 10^{-12} \times \frac{2 \times 10^{16}}{2 \times 10^{-6}} = -20.8 \times 10^{-9} = -0.0208 \times 10^{-6} \text{ A}$$

$$I_T = 1352 \times 10^{-9} - 0.0208 \times 10^{-6}$$

(9) given

$$D = 1.03 \text{ mm}$$

$$\frac{R}{l} = 0.0214 \Omega/\text{m}$$

$$\text{Concn of free } e = 8.4 \times 10^{28} \text{ electron}/\text{cm}^3$$

$$I = 2 \text{ A}$$

To find $V_d = ?$ $\mu_e = ?$

$$\rightarrow \frac{R}{l} = \frac{\rho}{A} = 0.0214 \Omega/\text{m}$$

$$\rho = 0.0214 \times A = 0.0214 \times \pi \times \frac{(1.03 \times 10^{-3})^2}{4}$$
$$= 0.0178 \times 10^{-6}$$

$$\sigma = \frac{1}{\rho} = \frac{10^6}{0.0178} = 56.08 \times 10^6$$

$$J = \sigma E$$

$$\frac{I}{A} = \sigma E = \frac{2 \times 4}{\pi \times (1.03 \times 10^{-3})^2} = 2.4 \times 10^6$$

$$\sigma E = 2.4 \times 10^6$$

$$E = \frac{2.4 \times 10^6}{56.08 \times 10^6} = 0.043 \text{ V/m}$$

$$V_d = \mu_e E \quad (1)$$

•

$$\sigma = q n$$

$$\mu = \frac{\sigma}{q n} = \frac{56.08 \times 10^6}{1.6 \times 10^{-19} \times 8.4 \times 10^{28} \times 10^6}$$

$$= \frac{56.08 \times 10^{25-28-6}}{1.6 \times 8.4}$$

$$= 4.17 \times 10^{-9} \text{ s/V}$$

$$V_d = 0.00417 \times 0.043 \times 10^6$$
$$= 1.79 \times 10^{-10} \text{ m/sec}$$

Ans 2

1. Given

n-type

$$\text{length} = 0.1 \mu\text{m}$$

$$\text{Area} = 0.05 \mu\text{m} \times 0.05 \mu\text{m}$$

$$\text{Vol} = 1\text{V}$$

$$N_D = 10^{17} \text{cm}^{-3}$$

$$I_D = ?$$

$$I_D = \mu n e A E$$

$$\text{assum } \mu = 1300$$

$$I_D = 1300 \times 10^{17} \times 1.6 \times 10^{-19} \times \frac{0.0025 \times 10^{-12} \times 1}{0.1 \times 10^{-6}}$$

$$= 13 \times 1.6 \times 0.0025 \times 10^{-5}$$

$$= 0.52 \times 10^{-6} \text{ A}$$

$$= 0.52 \mu\text{A} \quad \text{Ans}$$